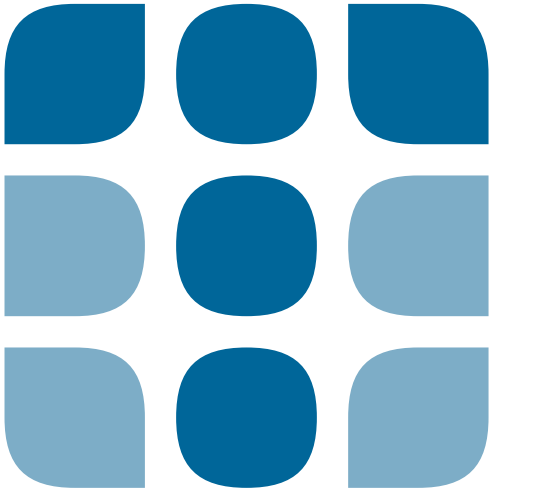


Sub 6 GHz versus mmWave Measurements in a Controlled High-Mobility Environment

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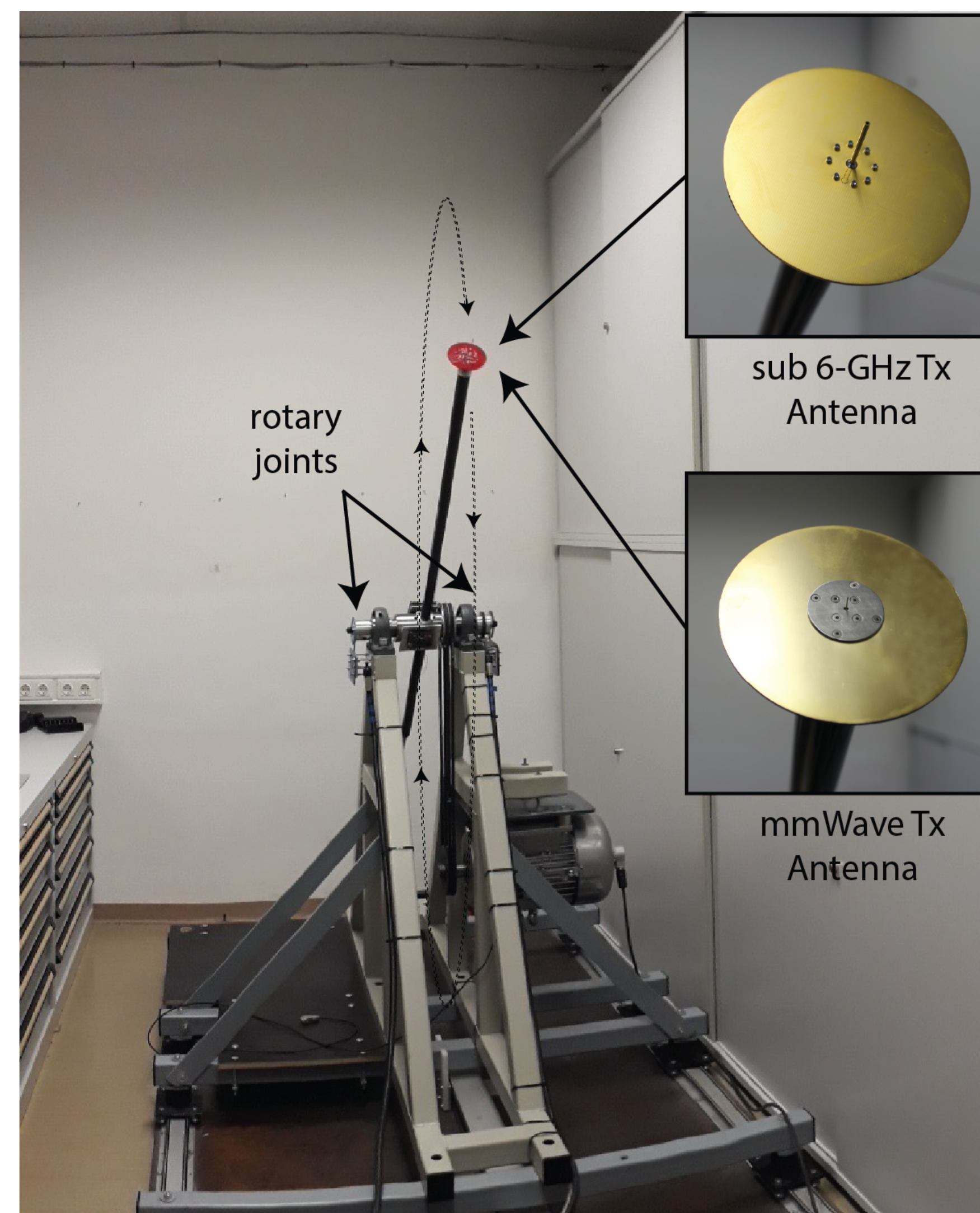
① Motivation

- ◇ high-mobility scenarios pose challenges for performance measurements
 - measurements cannot be easily repeated
 - expensive, complex and time-consuming
- ◇ comparing consecutive measurements in different bands is not straightforward

② Testbed Architecture

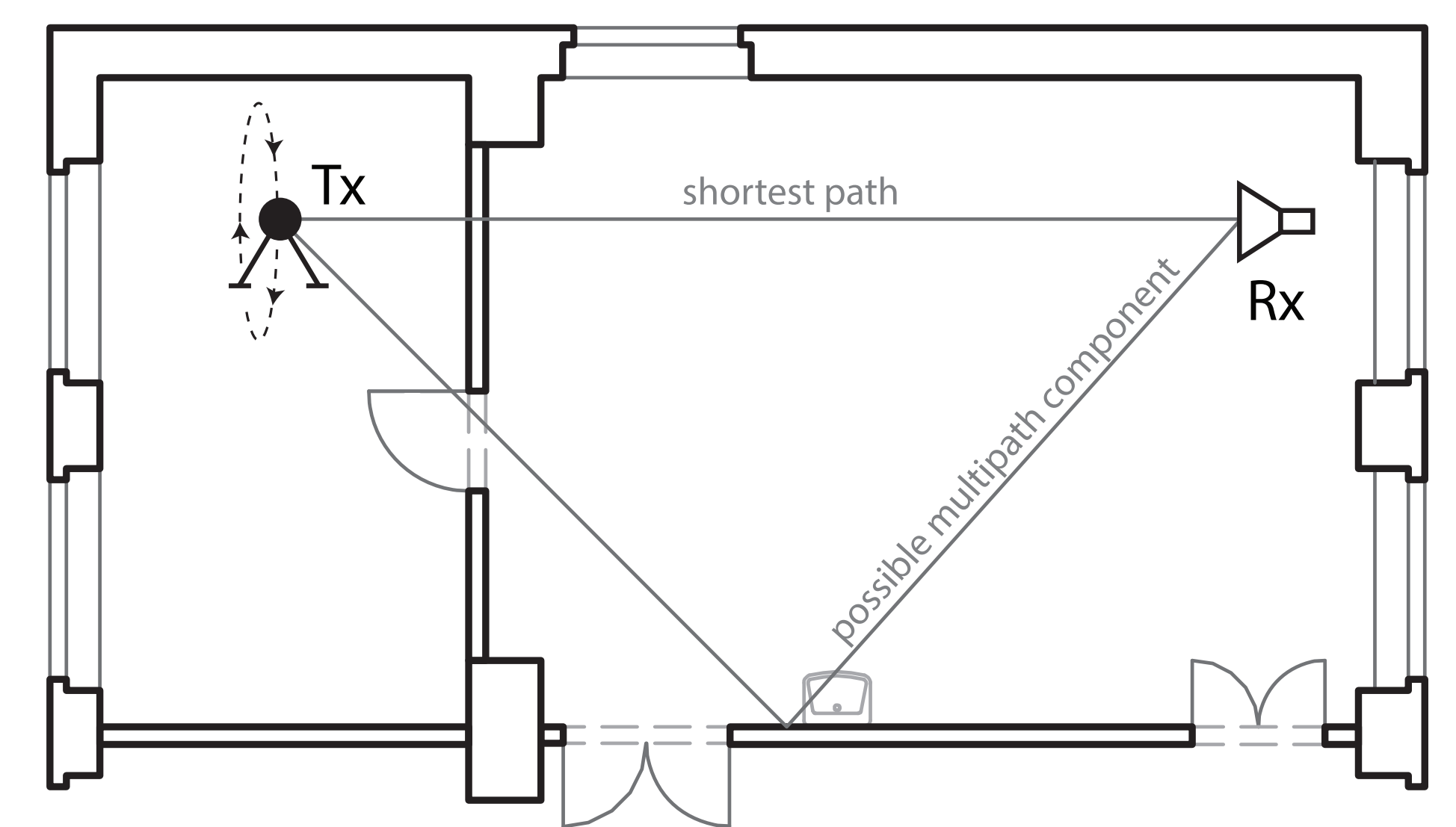
- ◇ rotary unit rotates an antenna around a central pivot with a constant speed of up to 400 km/h
- ◇ transmit signal fed through a rotary joint
- ◇ testbed setup allows for repeatable measurements in a controlled environment

③ Rotary Unit



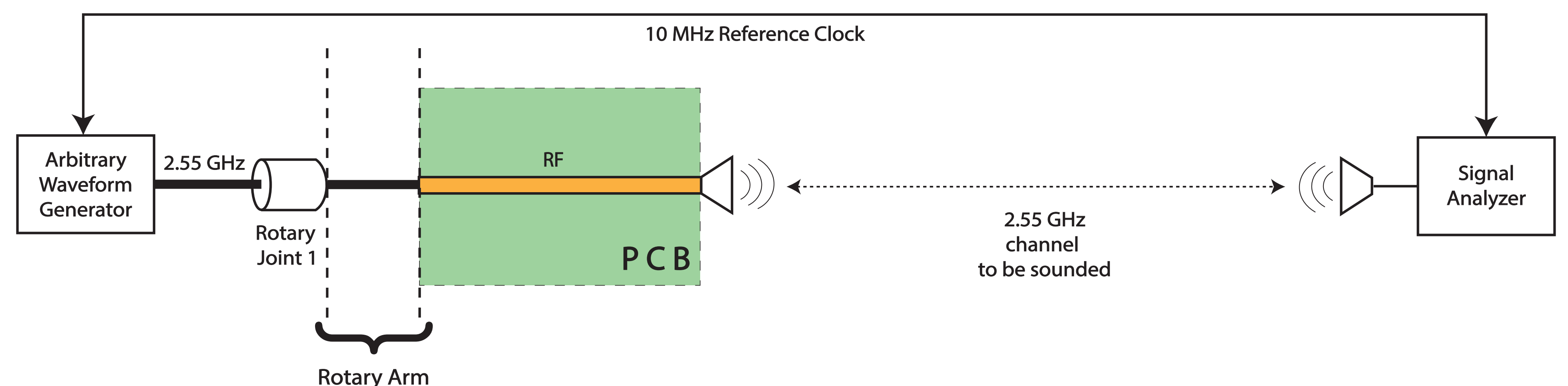
④ Measurement Environment

- ◇ testbed located in rich scattering laboratory environment
- ◇ rotating Tx uses an omnidirectional monopole antenna to enhance multi-path characteristics
- ◇ static Rx uses a directive horn antenna to obtain high gain



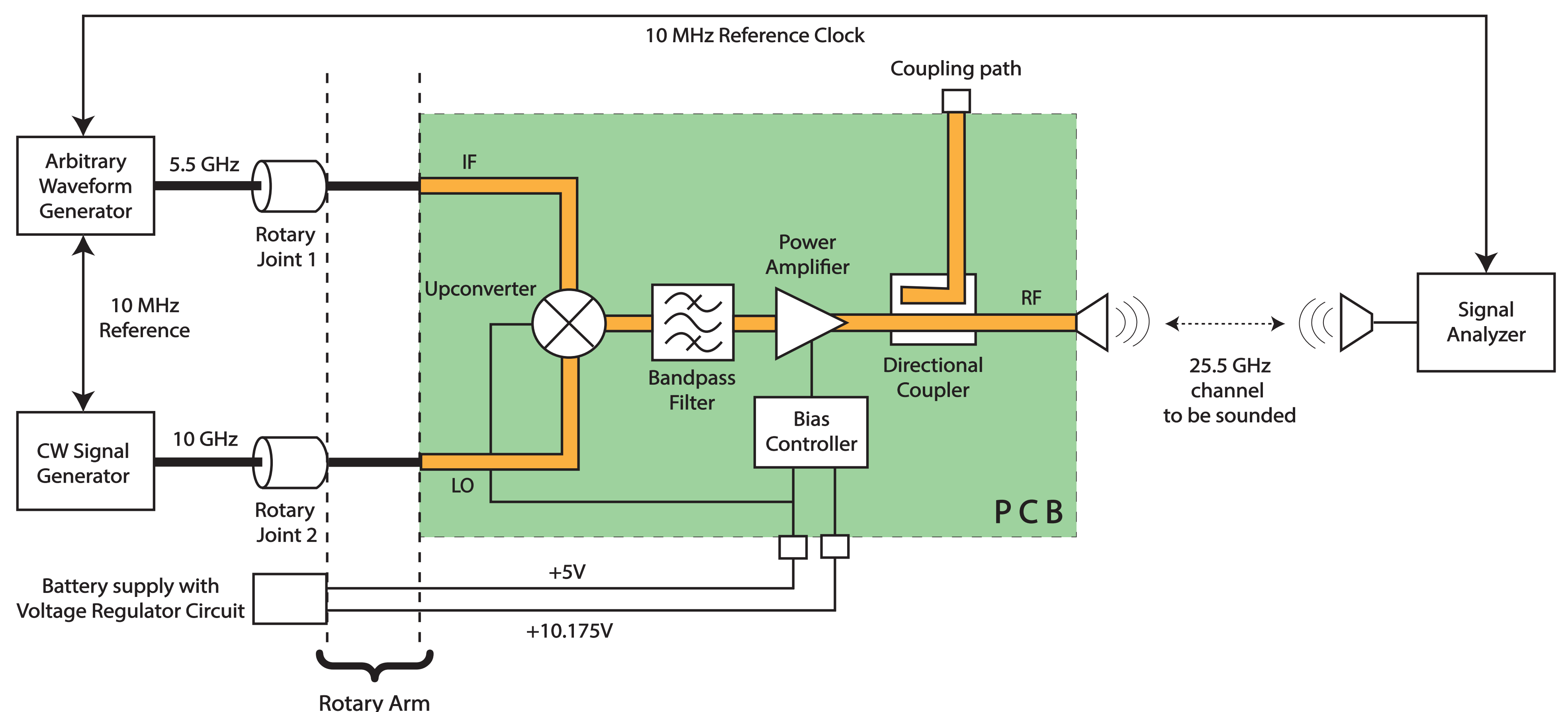
⑤ Sub 6 GHz Measurement Setup

- ◇ digital up-conversion to $f_{RF} = 2.55$ GHz by an Arbitrary Waveform Generator (AWG)
- ◇ up-converted signal transmitted directly to the end of the rotary arm through a rotary joint and cables
- ◇ signal radiated by an 2.55 GHz omnidirectional monopole antenna



⑥ mmWave Measurement Setup

- ◇ limitation: the rotary joint has upper-frequency limit of 18 GHz
- ◇ possible solution:
 - transmit 5.5 GHz IF signal and 10 GHz LO tone
 - up-conversion to RF range: $f_{RF} = 2f_{LO} + f_{IF}$
 - signal radiated by 25.5 GHz omnidirectional monopole antenna
- ◇ mmWave Tx designed using lightweight surface mount device (SMD) components
- ◇ additional SMD components
 - band-pass filter
 - power amplifier + active bias controller
 - directional coupler



⑦ Synchronization

- ◇ frequency synchronization: the AWG at the Tx and the Signal Analyzer at the Rx interconnected with a 10 MHz reference
- ◇ time synchronization: trigger a measurement at a precisely defined position of a rotary arm by using a rotational encoder

⑧ Summary

- ◇ measurement methodology for performance comparison of sub 6 GHz and mmWave high-mobility scenarios
- ◇ repeat measurements with equal channel conditions (hardly possible in real-world drive-by measurements)
- ◇ control environmental and system parameters in high-mobility environment
- ◇ conduct comparisons of measured real-world channels in terms of small scale and fast-fading